The community has also explored a wide range of domain-specific problems when it comes to LLM creativity, including logical-based problem-solving [13, 47, 71], physical and commonsense reasoning [68], creative writing [4, 10, 27, 35, 45, 58, 67], scientific discovery [1, 40, 63], response diversity in question answering [48, 73], and human-ai collaborative creative problem solving [8, 11, 50]. All these studies focus on LLM evaluation in a specific domain, each with its own evaluation philosophy and metrics, whereas our work aims at providing a holistic and comprehensive evaluation of the LLM's output for tasks in a variety of domains.

Automatic Creativity Evaluation Evaluating the creativity of machine-generated text has been a challenging task, and much of the work relies on human evaluation. However, due to the cost, human evaluation is challenging to scale and requires a lengthy wait time. To achieve scalable evaluation, researchers adapt two broad groups of evaluation methods: feature-based and generative-based. The former includes psycholinguistic features, such as arousal, valence score [49], lexical features, such as lexical diversity [55], and text embedding distances [57, 72]. The latter is mainly LLM-as-a-judge [42, 43, 66]. Recent work has shown promising potential of this method [75] as well as quantitative ways to measure how well LLM-Judge aligns with human judgment [9, 28]. Our evaluation framework integrates both feature-based and generative evaluators. For metrics that rely on LLM judges, we systematically validate their output against human annotations using the Alternative Annotator Test (see §4).

Task	Example
🧠 **Alternative Uses Test (AUT)** [22]	Create a list of creative alternative uses for a bottle.
🧠 **Divergent Association Task (DAT)** [6, 12]	Please write 10 nouns in English that are as irrelevant from each other as possible, in all meanings and uses of the words.
🧠 **Torrance Tests of Creative Thinking (TTCT)** [74]	What might be the consequences if humans suddenly lost the ability to sleep?
✍️ **Torrance Test of Creative Writing (TTCW)** [10]	Write a New Yorker-style story given the plot below. Make sure it is at least 2000 words. Plot: A woman experiences a disorienting night in a maternity ward...; Story:
✍️ **Creative Short Story** [35]	Come up with a novel and unique story that uses the required words in unconventional ways or settings. Use at most five sentences. The given words: petrol, diesel, and pump.
✍️ **Creativity Index** [45]	Please write a few paragraphs for a novel starting with the following prompt: "It's been years: Bailey clearly means him no harm and has managed to..."
💻 **NeoCoder** [47]	You are given a sequence of integers a of length $2n$. You have to split them into n pairs. Don't use HashMap, while loop.
💻 **Creative Math** [71]	Question: What is the largest power of 2 that is a divisor of 134 - 114? A.8 B.16 C.32 D.64 E.128; Reference Solutions 1: ... ; Reference Solutions 2: ...

Table 1: Tasks in CREATIVITYPRISM with examples. 🧠: divergent thinking, ✍️: creative writing, 💻: logical reasoning. More details and examples can be found in Appendix F.

3 CREATIVITYPRISM

Task Selection In designing CREATIVITYPRISM, we make sure it is both holistic and scalable. These two requirements are reflected on task selection: We examine all creativity-related tasks with publicly available data and executable, well-documented codebases We also further ensure the reliability of tasks that require LLM-Judge by conducting additional human annotation with trained researcher and LLM-Judge quality test to filter out tasks where no well-aligned LLM-Judge is available (see §4.2 for more details). Our task selection process result in eight tasks and seventeen metrics, spanning three domains: 🧠 divergent thinking, ✍️ creative writing, and 💻 logical reasoning, as shown in Figure 1. The divergent thinking domain consists of established psychology tasks, which were originally designed to assess human ability in coming up with diverse and alternative answers to given questions [6, 12, 22, 74]. The creative writing domain includes tasks that require models to produce short written pieces [10, 35, 45]. The logical reasoning domain includes coding [47]

and math task [71] to evaluate models' ability to generate creative solutions under strict, explicit reasoning constraints. Task input examples are in Table 1.

Three Dimensions of Creativity We go beyond simple combination of existing tasks by categorizing all metrics into three dimensions of creativity: *quality*, *novelty*, and *diversity*. This taxonomy is inspired by previous work in psychological creativity evaluation [2, 61]. We believe that a creative LLM should be able to generate *novel* and *diverse* responses with high *quality*. With this categorization, we evaluation of LLMs is not limited into one dimension, but in terms of different dimensions of cognitive capability, even if the tasks come from different domains. Detailed description of each dimension are following:

Quality includes metrics that evaluate how well the generated content fulfills the task's functionality. For example, in NeoCoder, the quality of generated code is measured by the success of execution and coding task completion; in TTCT, quality is measured by the amount of detail provided. We believe quality is critical for creativity because, without quality, random words and sentences would be very creative in terms of novelty and diversity, but do not convey any meaning.

Novelty includes metrics that evaluate how rare the generated content is compared to existing or commonly seen content. For example, in NeoCoder and Creative Math, novelty involves coming up with solutions that are different from the reference solutions; in AUT, novelty involves different use of the tool compared to ordinary uses; in Creativity Index, novelty is measured by the normalized n-gram overlaps between model-generated text and the traceable part of the training corpus.

Diversity includes metrics that evaluate how much the LLM-generated content differs in different passes. For example, in TTCT, diversity is reflected in the flexibility score, the ability to produce varied solutions to given questions; for the DAT task, diversity involves the semantic difference among the LLM-generated nouns.

4 Experiments

To holistically evaluate machine creativity, we evaluated 17 SoTA LLMs across eight tasks, reporting both task-specific metrics and an aggregated creativity score using the three-dimensional framework. In this section, we introduce the inference setups, where LLMs are prompted to generate creative responses according to corresponding task requirements (§4.1); then we describe the evaluation process, including score aggregation (§4.2) and how we use LLM-as-a-Judge for evaluation (§4.2).

4.1 Inference

For all the tasks in CREATIVITYPRISM, we collect the original datasets. Unless otherwise specified, all the data processing is done according to the original papers. We include 17 LLMs in total, including open-source models from Mistral [36, 37], Qwen [33, 59], OLMo [25], Llama [24], and the Deepseek [15, 16] family, and proprietary LLMs from GPT [53], Claude [3], and the Gemini [23, 39] family. For open-sourced LLMs, we use `vLLM(v0.7.2)` [41] to run all experiments. For proprietary LLMs, we use API access from the corresponding company. Inference time parameters vary depending on the task. More inference configurations are in Appendix F.

4.2 Evaluation

Aggregated Creativity Scoring To capture overall performance within each dimension, we aggregate all metrics in that dimension to produce quality, novelty, and diversity scores. The score aggregation calculation involves two steps: first, every evaluation metric is min-max normalized to a 0–1 scale based on the minimum and maximum possible scores for the task. Second, the normalized metrics are aggregated into quality, novelty, and diversity scores for each LLM (as categorized in Figure 1) by averaging all metrics within each dimension. To ensure equal weighting across tasks, we first average the normalized scores within each task before computing dimension scores, this prevents tasks with multiple metrics in one dimension (e.g., TTCW has three novelty metrics) from disproportionately influencing the final score. More details about score aggregation can be found in Appendix C. We also provide an "overall" creativity score (shown in Table 3) by averaging across the three dimensions to facilitate holistic model comparison. Note that this "overall" score is only to facilitate model comparison. The relative importance of quality, novelty, and diversity varies

depending on the application. We suggest future researchers choose from these three separate scores based on their specific research goals.

Task	Fleiss Kappa	Judge-LLM	Judge Quality
AUT	0.650	GPT4.1	✓pass
NeoCoder	0.471	GPT4.1	✓pass
TTCW (Originality in Theme/Content)	0.660	Qwen2.5-72B	✓pass
TTCW (Originality in Thought)	0.400	Qwen2.5-72B	✓pass
TTCW (Originality in Form)	0.410	Qwen2.5-72B	✓pass
TTCW (Perspective & Voice Flexibility)	0.440	Qwen2.5-72B	✓pass
TTCW (Narrative Ending)	0.480	Qwen2.5-72B	✓pass
TTCT (Fluency)	0.432	GPT4.1, Qwen2.5-72B	✓pass
TTCT (Flexibility)	0.423	GPT4.1, Qwen2.5-72B	✓pass
TTCT (Elaboration)	0.445	GPT4.1, Qwen2.5-72B	✓pass
CreativeMath (Novelty)	0.450	Claude3-Sonnet, GPT4.1, Gemini2.0-Flash	✓pass
CreativeMath (Correctness)	-	Claude3-Sonnet	*accuracy: 0.91*

Table 2: Inter-annotator agreement and Judge-LLM quality. "✓pass" refers to passing the Alternative Annotator Test; "accuracy" is the accuracy compared to ground truth.

LLM-as-a-Judge Reliability In CREATIVITYPRISM, the evaluation of five tasks (AUT, TTCW, CreativeMath, TTCT, and NeoCoder) involves using LLM as part of the automatic evaluation procedure. To ensure the reliability of the LLM judges, we verify two requirements for every metric in these tasks. First, we collect human annotations and compute inter-annotator agreement, confirming that the task is well-defined and that annotators can reach reasonable agreement on the evaluation outcomes. Second, we follow Calderon et al. [9] and test if our LLM judges can replace human annotators by comparing human annotations with the judgment from LLMs.

For human inter-annotator agreement, we randomly sample a small subset of data points from the output of inference models and have annotators rate the inference output based on the task rubrics. Annotators are all researchers familiar with the task. Three annotators annotate each data point, and the inter-annotator agreement is measured by Fleiss Kappa [20], or Weighted Fleiss Kappa [14], if the labels are in Likert scale. Given the subjective nature of creativity-related tasks, we keep tasks where the Fleiss Kappa measurements among annotators are higher than 0.4, which implies moderate agreement. See detailed agreements in Table 2. Note that the Judge quality of CreativeMath (Correctness) is evaluated by accuracy (compared to human judgment) because the correctness of a solution to a math problem is objective. We simply have annotators verify the correctness of each solution for the sampled questions.

After verifying human agreement, we validate our choice of LLM (referred to as "Judge-LLM"). For the objective task, which only includes Correctness in CreativeMath, we simply calculate the accuracy of our Judge-LLM compared to human judgments. For the subjective tasks, we employ the Alternative Annotator Test [9] to statistically justify replacing human annotators. This method adopts a leave-one-out strategy to validate the LLM as a reliable substitute: for each human annotator, we test whether the Judge-LLM aligns closer to the remaining human consensus than the excluded human does (subject to a quality margin ϵ). If the LLM aligns better in more than half the cases (winning rate > 0.5), we consider it a viable replacement.

Note that the choice of backbone LLM clearly has an impact on whether or not the Judge-LLM passes this test. We follow the following principle to choose Judge-LLM: we use Qwen2.5-72B [59] as the default backbone LLM due to its open-source nature; if it does not pass the replacement test, we use GPT-4.1 [53]; if GPT-4.1 does not pass the test, we first try taking the average of GPT-4.1 and Qwen2.5-72B; lastly, we will use the Judge-LLM setup proposed by the original paper that introduced the task. All the Judge-LLM we included in CREATIVITYPRISM pass the Alternative Annotator Test when compared to human annotations. [2]

[2] We exclude sub-tasks and metrics that fail to meet our reliability requirements (Fleiss Kappa > 0.4 and passing the Alternative Annotator Test), though we minimize such modifications whenever possible (details in Appendix D).